UNIVERSITY OF AUCKLAND
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SCIENCE

COMPSCI 714
Lecture 12: Transfer Learning II

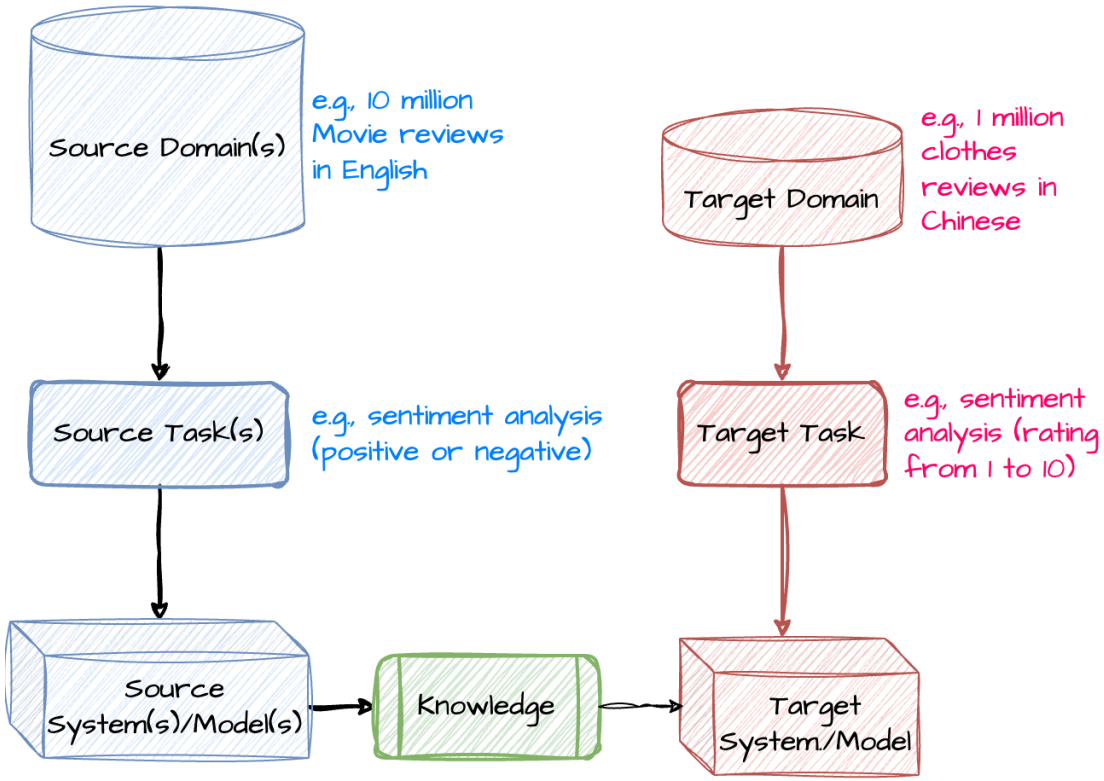
Transfer Learning Methods

Instructor: Qian Liu

Transfer Learning -- Overview

Definition of Transfer Learning:

- Given a **source** domain $\mathcal{D}_s$ with corresponding learning task $\mathcal{T}_s$,
- a **target** domain $\mathcal{D}_t$ with corresponding learning task $\mathcal{T}_t$,
- **transfer learning** is the process improving the target predictive function $f_t(\cdot)$ in $\mathcal{D}_t$ by using the knowledge in $\mathcal{D}_s$ and $\mathcal{T}_s$, where $\mathcal{D}_s \neq \mathcal{D}_t$ or $\mathcal{T}_s \neq \mathcal{T}_t$



Transfer Learning -- Overview

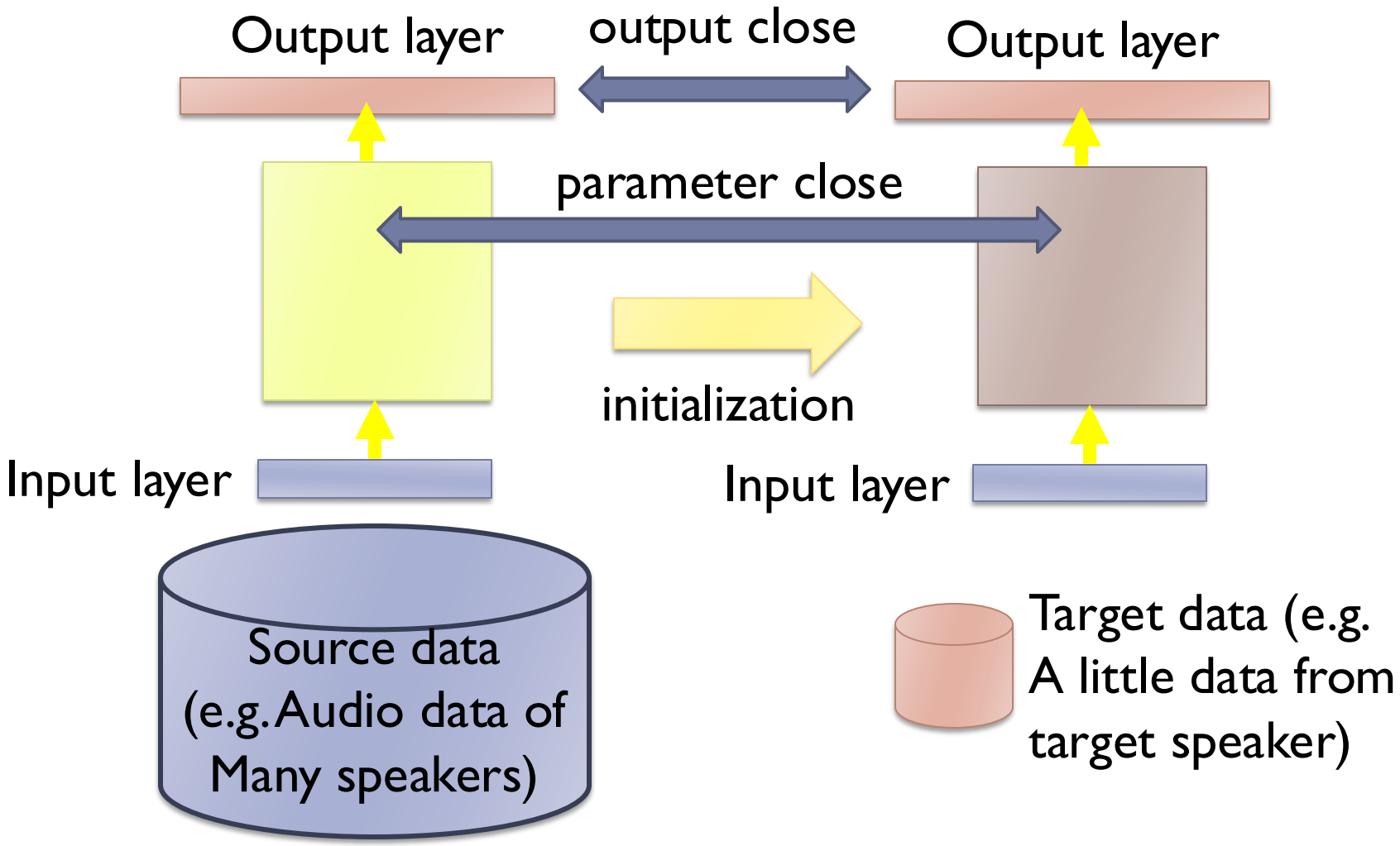
		Source Data (not directly related to the task)	
		labelled	unlabeled
Target Data	labelled	Model Fine-tuning	
	unlabeled		

Model Fine-tuning

- Task description
  - Source data: (x^s, y^s) ⇐ A large amount
  - Target data: (x^t, y^t) ⇐ Very little

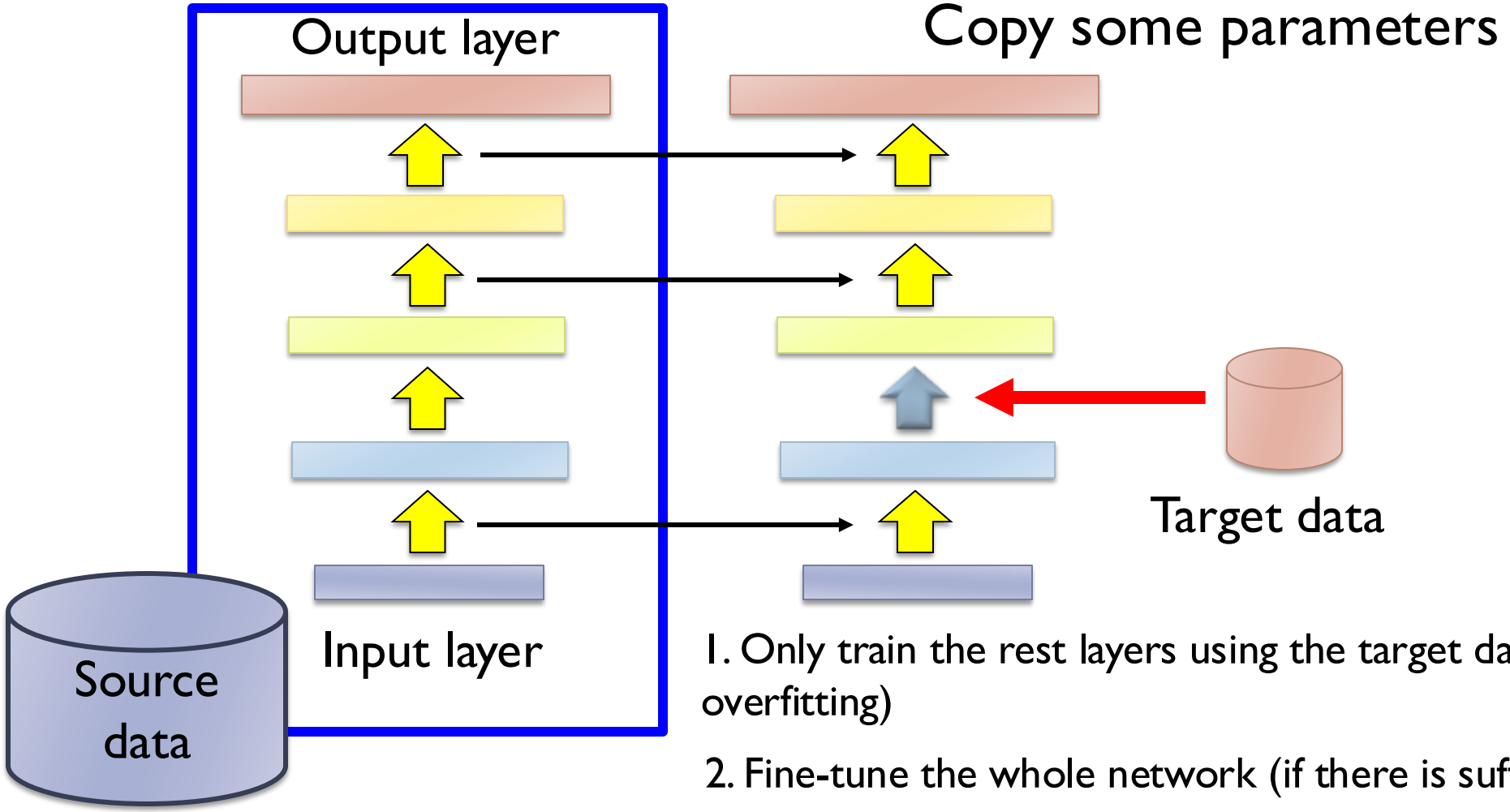
- Example: (supervised) speaker adaption
  - Target data: audio data and its transcriptions of specific user (e.g., we only have 3 examples from the specific user)
  - Source data: audio data and transcriptions from many speakers (e.g., we can collect large amount data from different speakers, even over 10K hours)
- Idea: training a model by source data, then fine-tune the model by target data
  - Use the trained model from source data as the initial model, and continue train the model with the target data
  - Challenge: only limited target data, so be careful about overfitting

Conservation Learning



- Ensure the difference between the new and old models is not too large to avoid overfitting.
- Constrains:
  - With the same input data, the new model output and the old model output should be close
  - The parameters of new model and old model should be close, e.g., measured through metrics like L2 norm

Layer Transfer



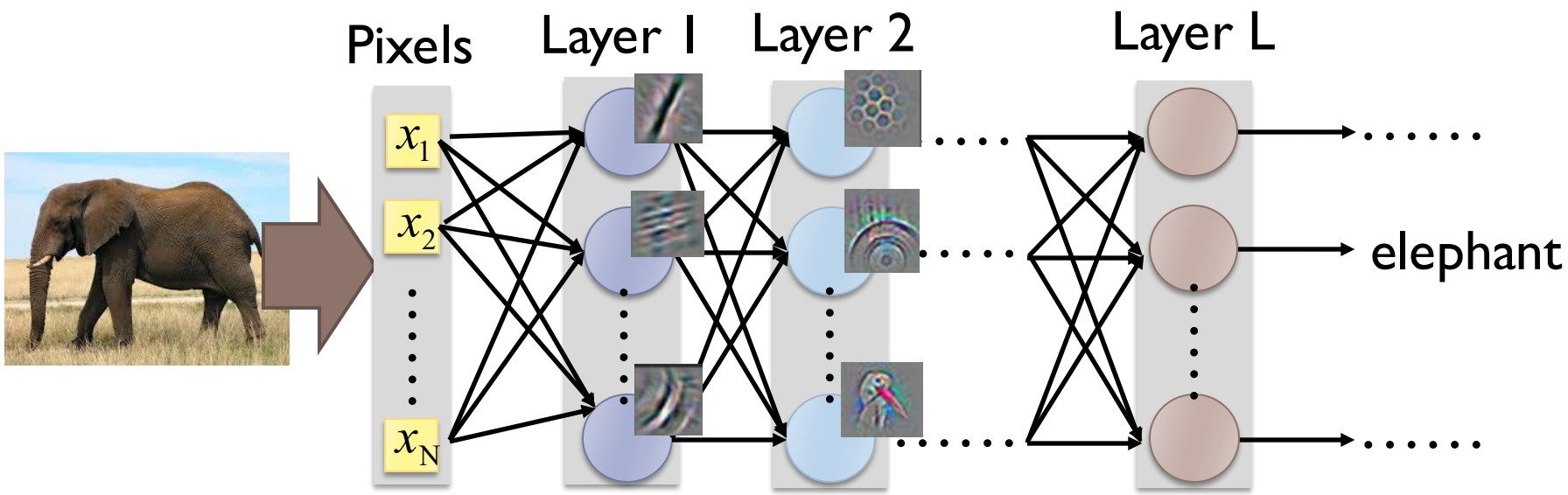
1. Only train the rest layers using the target data (prevent overfitting)

2. Fine-tune the whole network (if there is sufficient data)

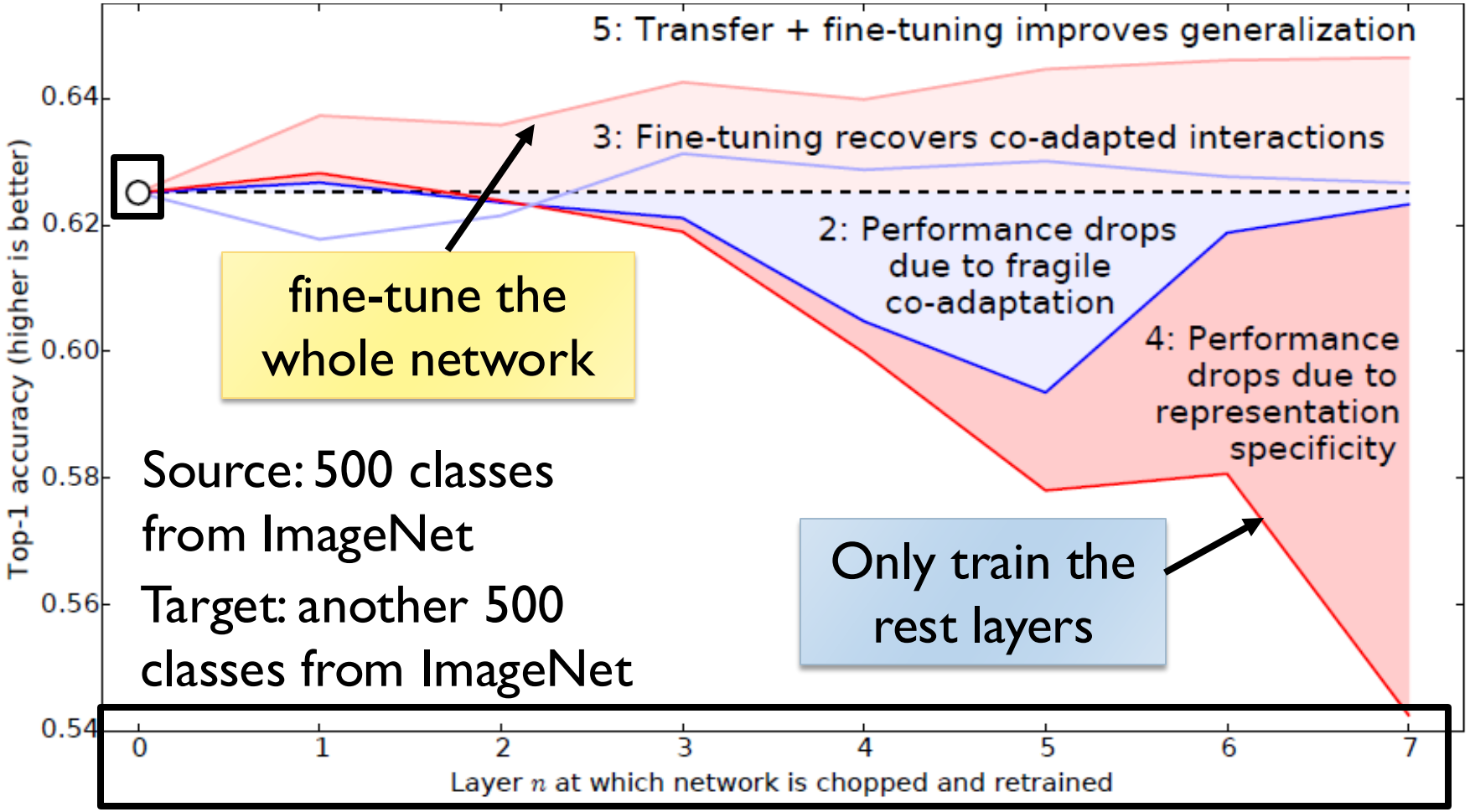
Layer Transfer

- Which layer can be transferred (copied)?
  - Speech: usually copy the last few layers
  - Image: usually copy the first few layers

Different tasks have different answers



Layer Transfer



Jason Yosinski, Jeff Clune, Yoshua Bengio, Hod Lipson, "How transferable are features in deep neural networks?", NIPS, 2014

Transfer Learning--Overview

		Source Data (not directly related to the task)	
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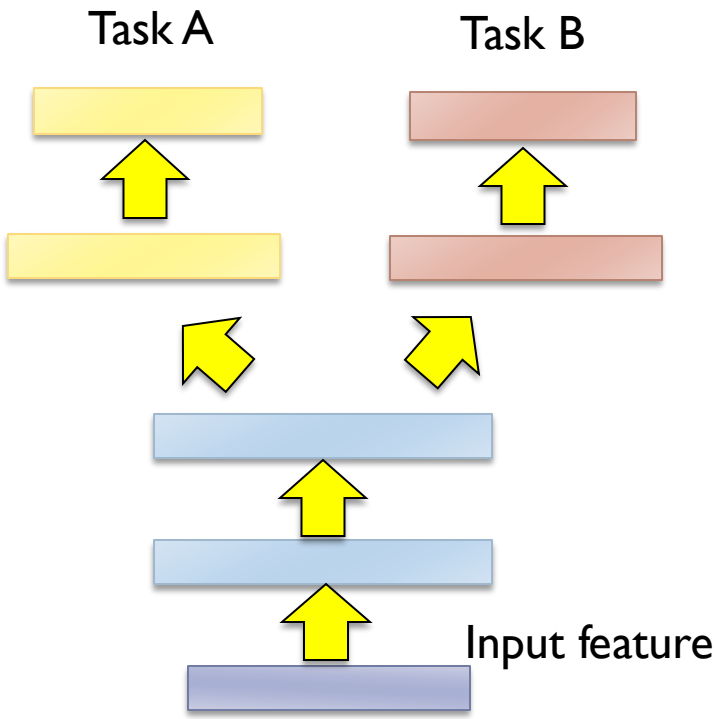
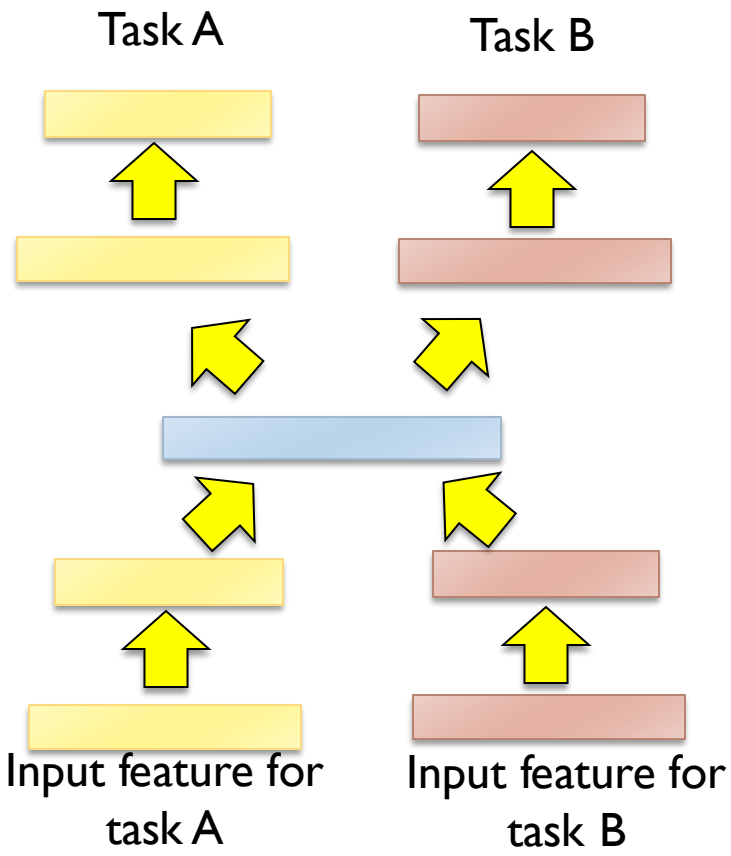
	unlabeled		

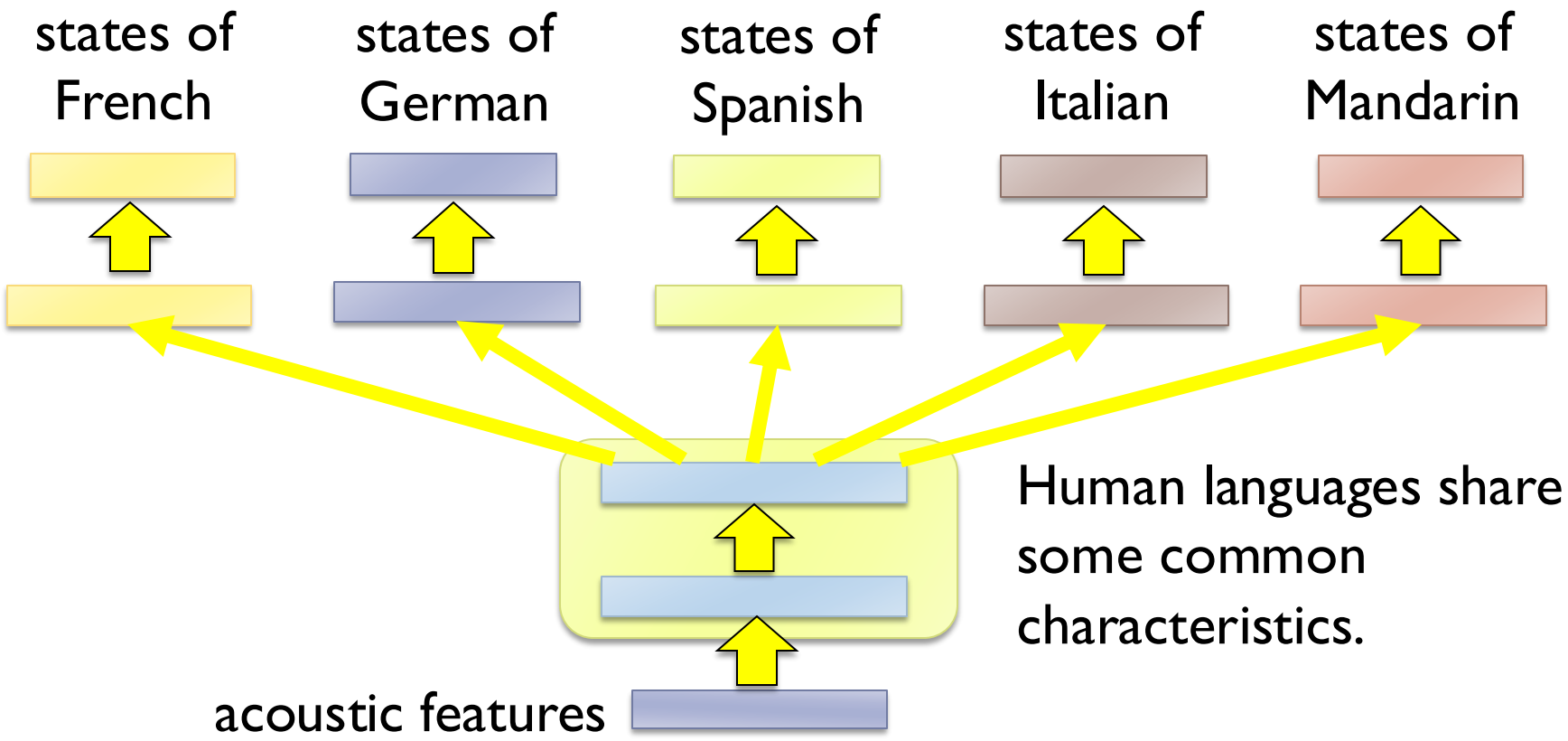
Multitask Learning

- The multi-layer structure makes NN suitable for multitask learning





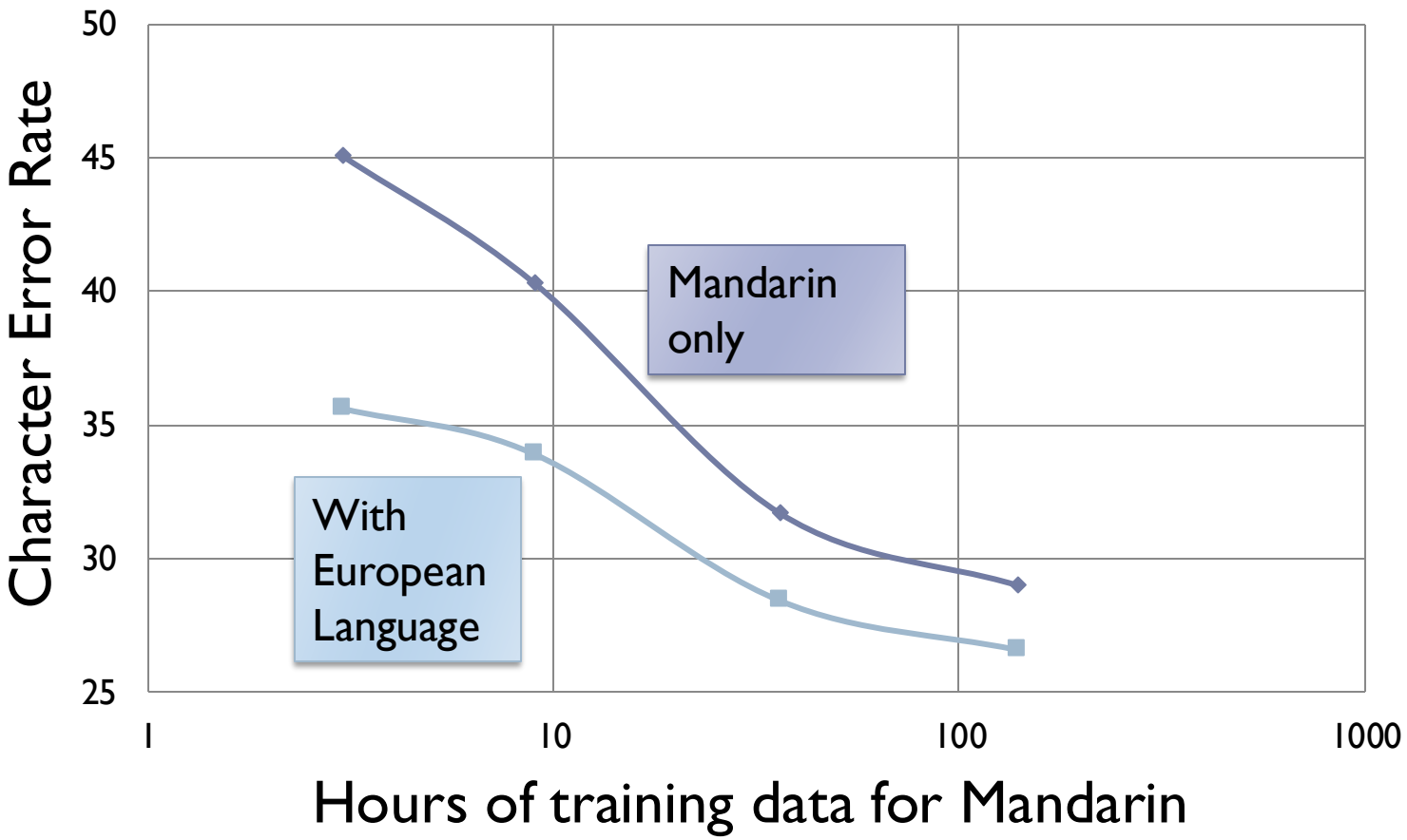
Multitask Learning--Multilingual Speech Recognition



Human languages share some common characteristics.

Similar idea in translation: Daxiang Dong, Hua Wu, Wei He, Dianhai Yu and Haifeng Wang, "Multi-task learning for multiple language translation.", ACL 2015

Multitask Learning - Multilingual



Huang, Jui-Ting, et al. "Cross-language knowledge transfer using multilingual deep neural network with shared hidden layers." ICASSP, 2013

Transfer Learning--Overview

		Source Data (not directly related to the task)	
		labelled	unlabeled
Target Data	labelled	Model Fine-tuning Multitask Learning	
	unlabeled	Domain-adversarial training	

Task Description

- Source data: (x^s, y^s) → Training data
- Target data: (x^t) → Testing data

Same task,
Different domains

MNIST

SOURCE

with label

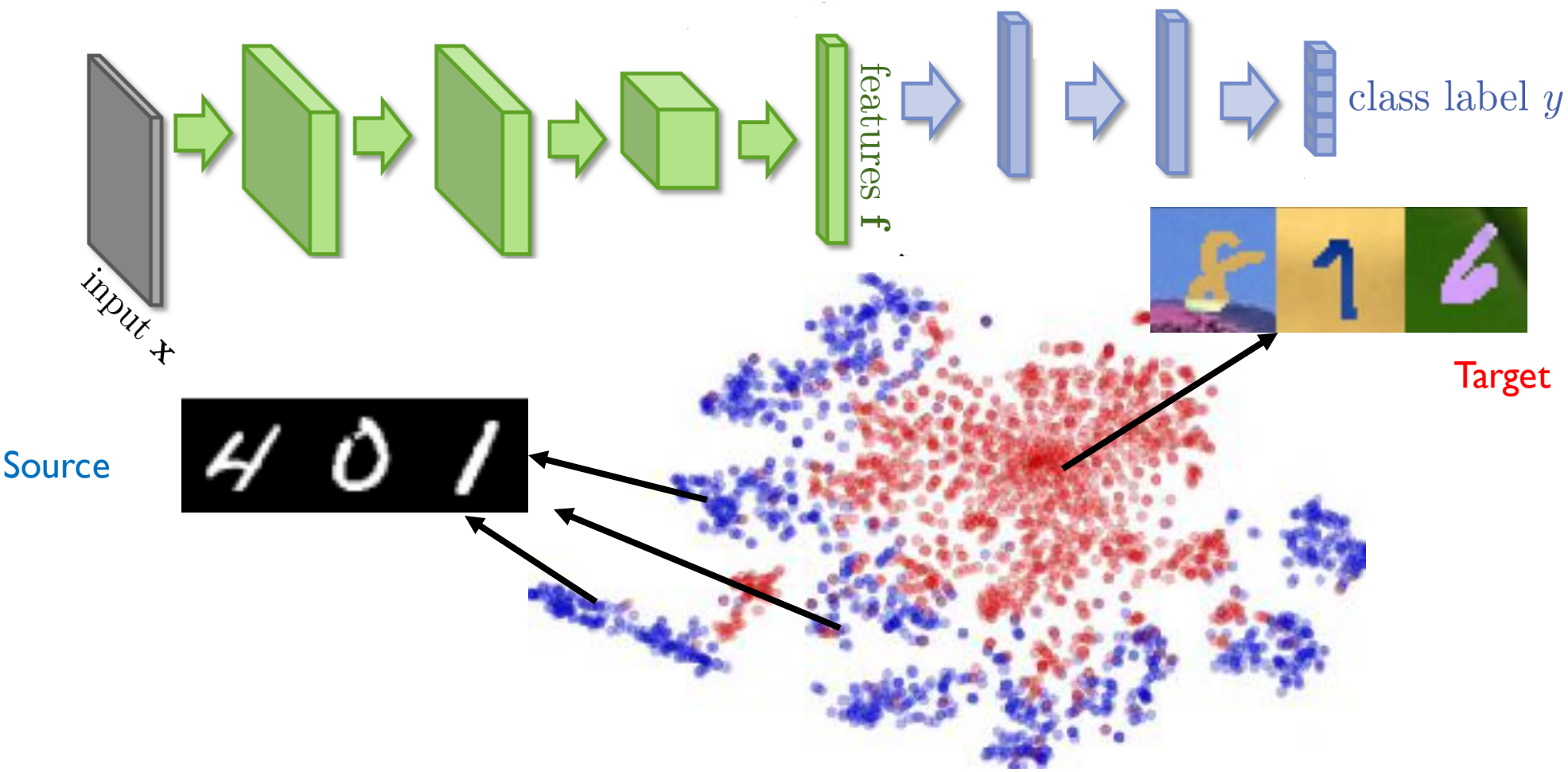
TARGET

without label

MNIST-M

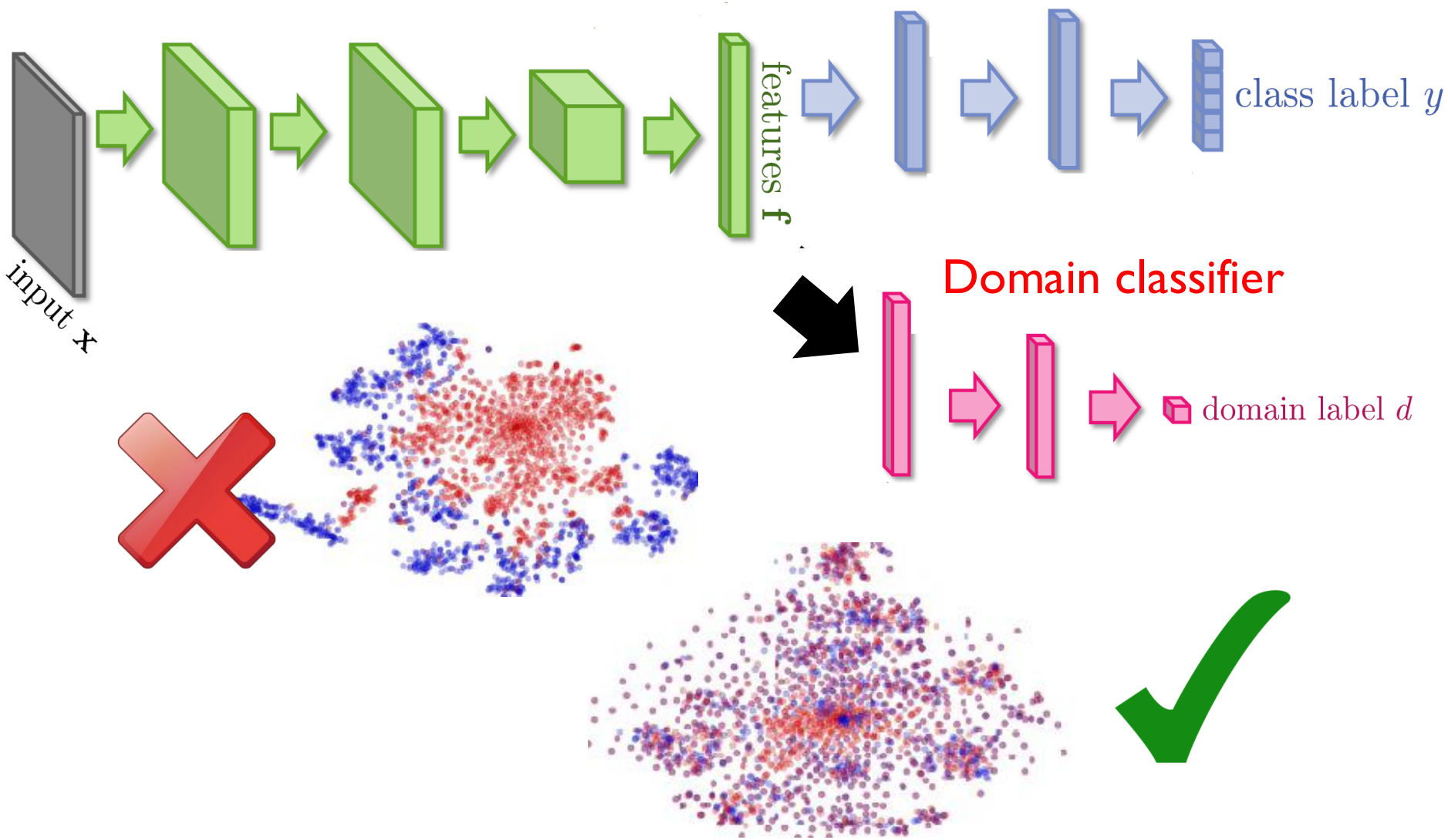
The MINIST dataset is a widely used dataset for handwritten digit recognition

Domain-adversarial training



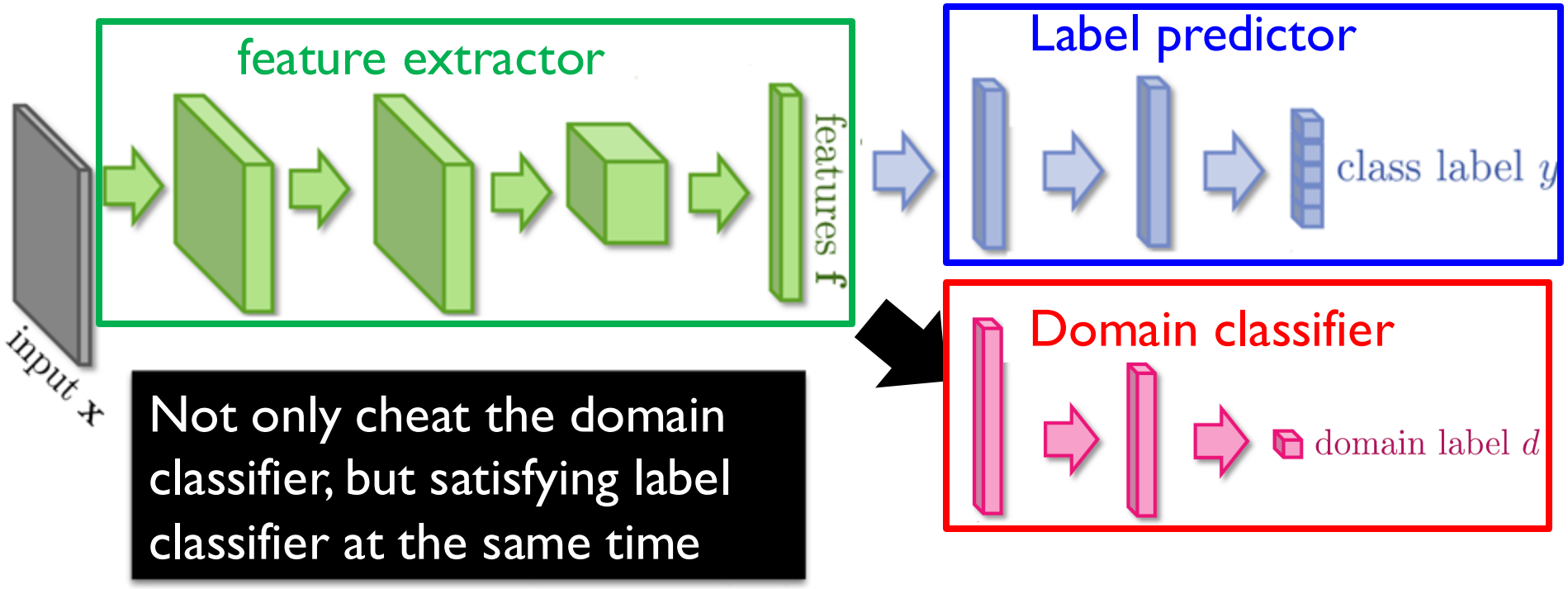
t-SNE (t-distributed Stochastic Neighbor Embedding) is an unsupervised non-linear dimensionality reduction technique for data exploration and visualizing high-dimensional data.

Domain-adversarial training



Domain-adversarial training

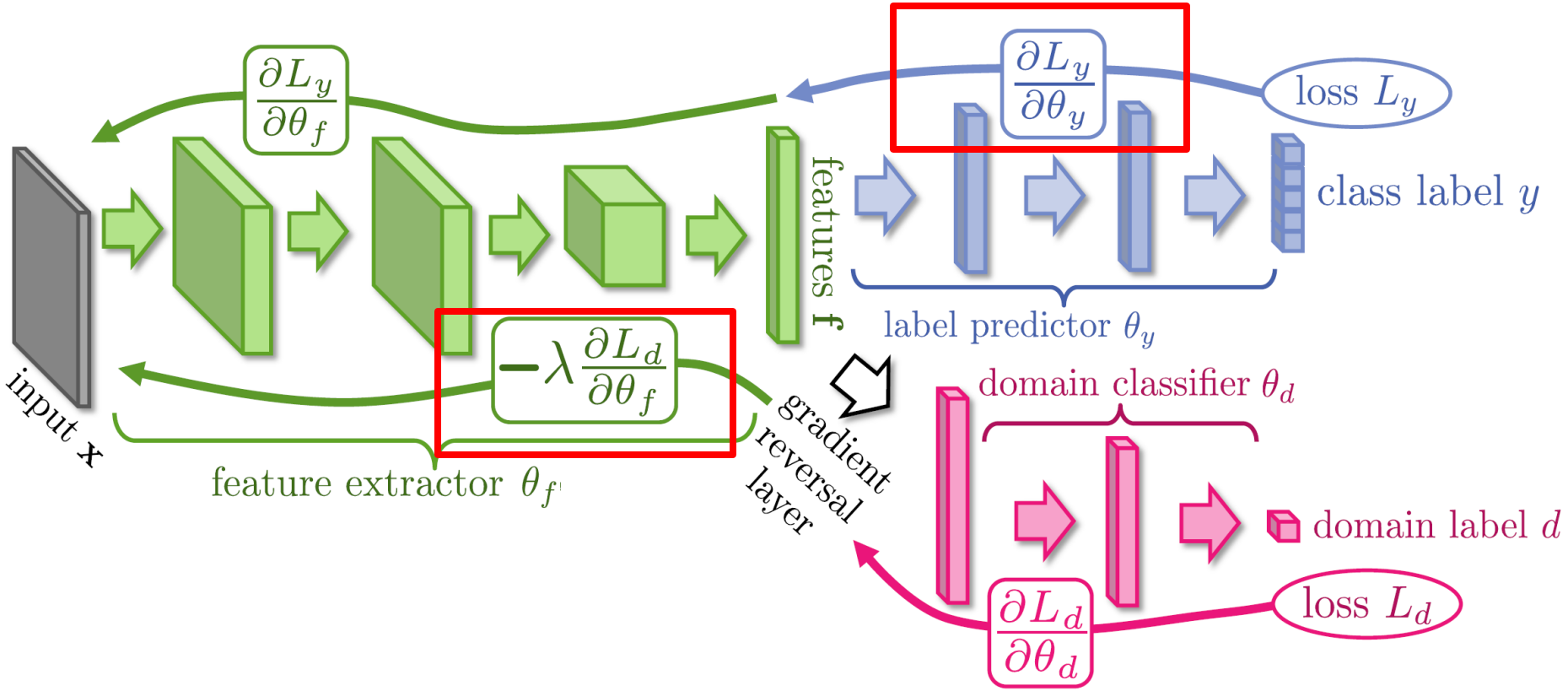
Maximize label classification accuracy + minimize domain classification accuracy



Not only cheat the domain classifier, but satisfying label classifier at the same time

This is a big network, but different parts have different goals.

Domain-adversarial training



Yaroslav Ganin, Victor Lempitsky, Unsupervised Domain Adaptation by Backpropagation, ICML, 2015

Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, Domain-Adversarial Training of Neural Networks, JMLR, 2016

Transfer Learning--Overview

		Source Data (not directly related to the task)	
		labelled	unlabeled
Target Data	labelled	Model Fine-tuning Multitask Learning	
	unlabeled	Domain-adversarial training Zero/Few-shot learning	

Task Description

- Source data: (x^s, y^s) → Training data
- Target data: (x^t) → Testing data

Different tasks

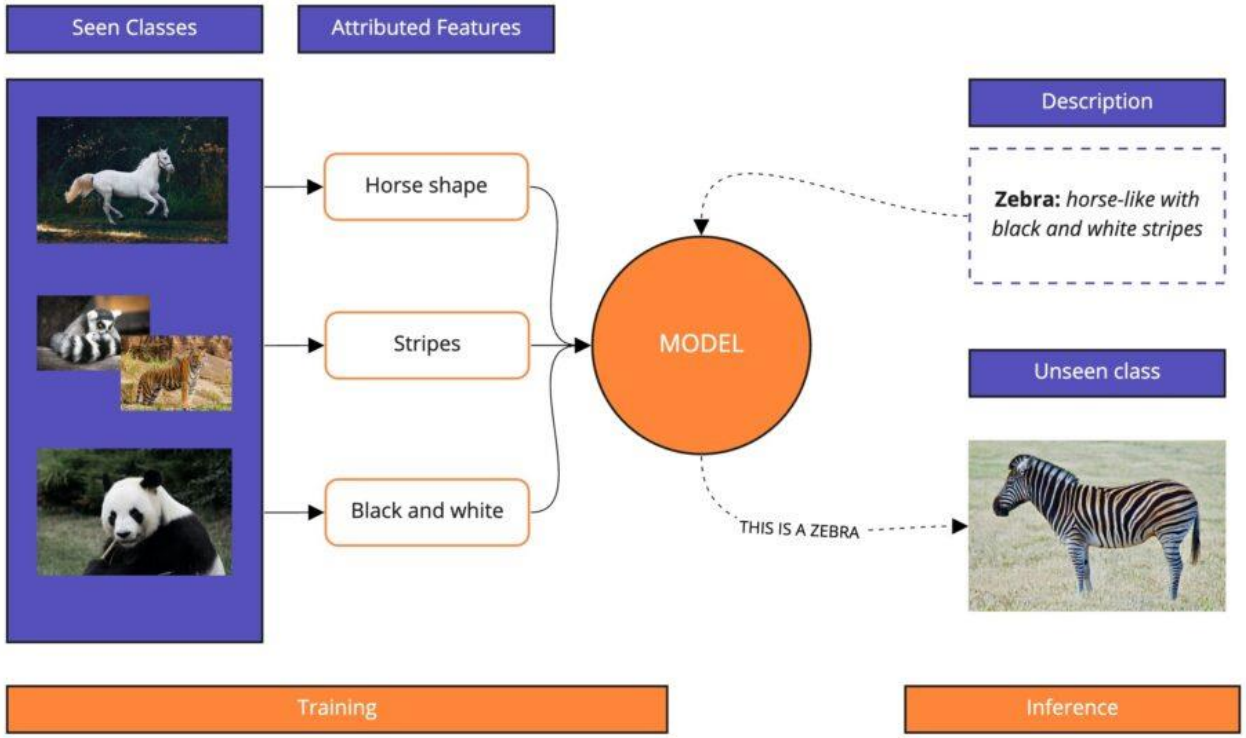
x^s:

y^s: cat dog

x^t:

Zero-shot Learning

- Example: classify images without having seen any samples from those categories during training.
- The goal is to generalize to new, unseen categories based on learned relationships and features.



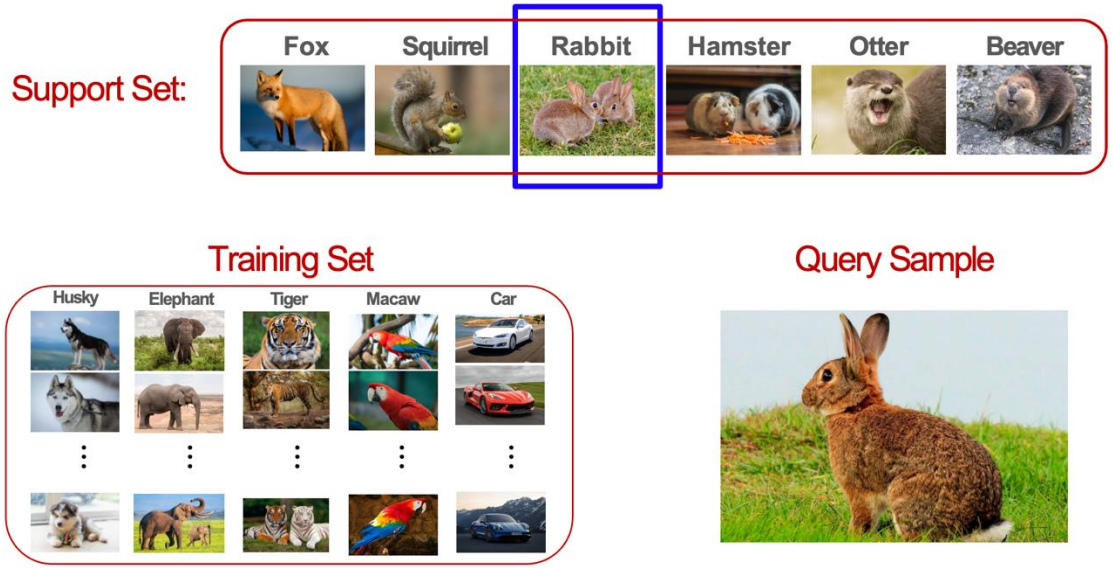
For example, someone who has seen a horse, but doesn't know *zebras* could likely recognize one knowing that a zebra looks like a *horse with back and white stripes*. Zero-shot learning assumes a semantic relationship between the seen and unseen classes.

Fig from https://modulai.io/blog/zero-shot-learning-in-nlp/

What is Zero-shot Learning

Few-shot Learning

- Example -- the model has access to a very limited number of samples during the training phase (typically a few).
- Utilize the limited samples and enhance the model's generalization ability (e.g., meta-learning or data augmentation)



Training Set, Support Set, and Query

- k-way: the support set has k classes.
- n-shot: every class has n samples.

Zero-shot Learning

- Instead of predicting the label, learning similarity!
  - First, a similarity function is trained from large-scale training dataset.
  - Then, apply the similarity function for prediction:

Bulldog

$\mathbf{x}_1$

Bulldog

$\mathbf{x}_2$

Fox

$\mathbf{x}_3$

- Compare the query with every sample in the support set.
- Find the sample with the highest similarity score.

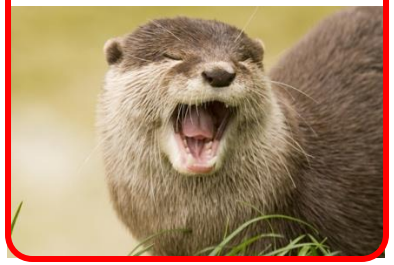
- Learn a similarity function: $\mathrm{sim}(\mathbf{x}, \mathbf{x}')$.
- Ideally, $\mathrm{sim}(\mathbf{x}_1, \mathbf{x}_2) = 1$, $\mathrm{sim}(\mathbf{x}_1, \mathbf{x}_3) = 0$, and $\mathrm{sim}(\mathbf{x}_2, \mathbf{x}_3) = 0$.

Learn a Similarity Function

What is in the image?

Query:

sim = 0.2	sim = 0.1	sim = 0.03	sim = 0.05	sim = 0.7	sim = 0.5
Greyhound	**Bulldog**	**Armadillo**	**Pangolin**	**Otter**	**Beaver**



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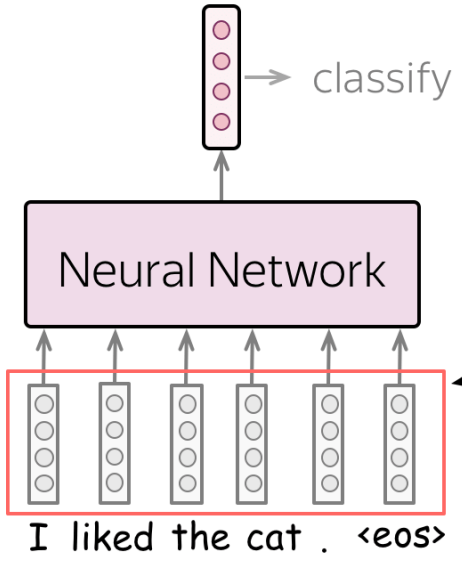
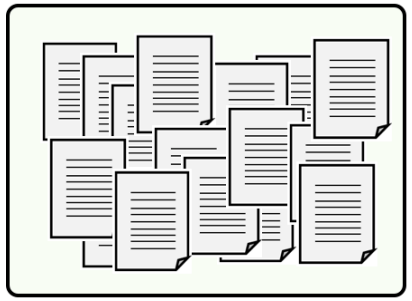
Target Data	labelled	Model Fine-tuning Multitask Learning	Self-taught Learning Rajat Raina , Alexis Battle , Honglak Lee , Benjamin Packer , Andrew Y. Ng, Self-taught learning: transfer learning from unlabeled data, ICML, 2007
	unlabeled	Domain-adversarial training Zero/Few-shot learning	Self-taught Clustering Wenyuan Dai, Qiang Yang, Gui-Rong Xue, Yong Yu, "Self-taught clustering", ICML 2008

Transfer Learning with Unlabeled Data

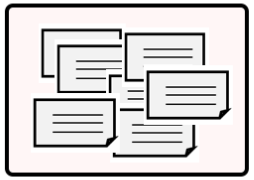
- Use the unlabeled source domain data to improve the performance of targe domain tasks.
- NLP Examples:
  - Word embeddings, Word2Vec, GloVe
  - Pre-trained language models, e.g., BERT, GPT





Training data for word embeddings (unlabeled)

- Huge diverse corpus (e.g., Wikipedia)
- Domain: general



Training data for text classification (labeled)

- Not huge, or not diverse, or both
- Domain: task-specific

Input word embeddings:

- Train from scratch
- Take pretrained (Word2Vec, GloVe)
- Initialize with pretrained, then fine-tune

Training on unlabeled source data;
Fine-tuning on labeled target data

https://lena-voita.github.io/nlp_course/word_embeddings.html

Word Embedding

- We don't have labeled data in source domain ☹
  - Let's use the unlabeled data! we can download millions of documents from web.
- Using the unlabeled data, how to define the training task?
  - Language Modeling!

S = Where are we going

Previous words (Context)

Word being predicted

P(S) = P(Where) x P(are | Where) x P(we | Where are) x P(going | Where are we)

what is language modeling?

Language modeling is the task of predicting the next word (or character, or token) in a sequence of text based on the words that came before it. It lies at the core of many natural language processing (NLP) tasks and is fundamental to systems like ChatGPT, machine translation, speech recognition, and text generation.

In more detail:

A **language model** learns the probability distribution of word sequences. For example:

- Given the input: *"The cat sat on the"*, a language model might predict:
  - `"mat"` with probability 0.7
  - `"sofa"` with probability 0.2
  - `"roof"` with probability 0.05
  - etc.

Word Embedding

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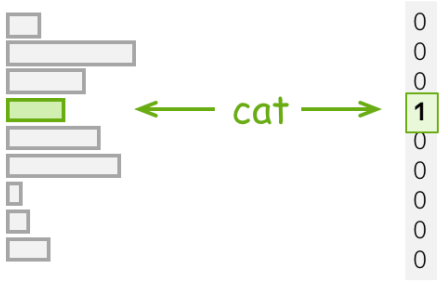
we want the model to predict this

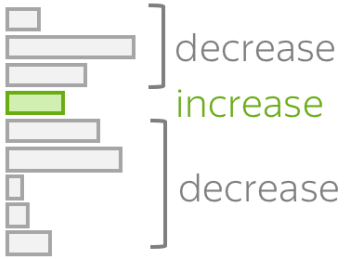
Training example: I saw a cat on a mat <eos>

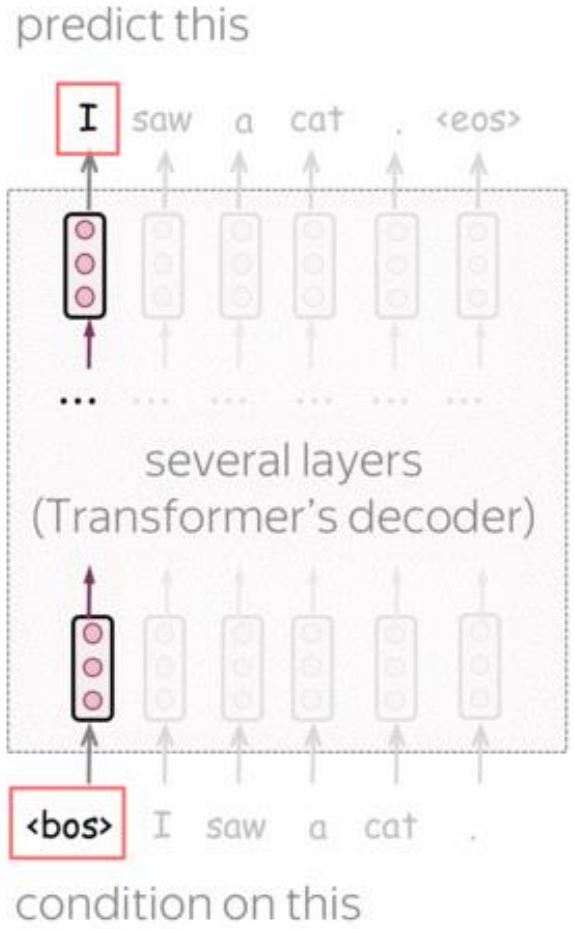
Model prediction: p(* | I saw a)

Target



Loss = -log (p(cat)) → min





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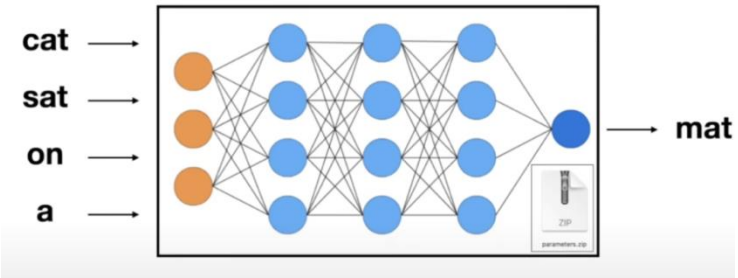
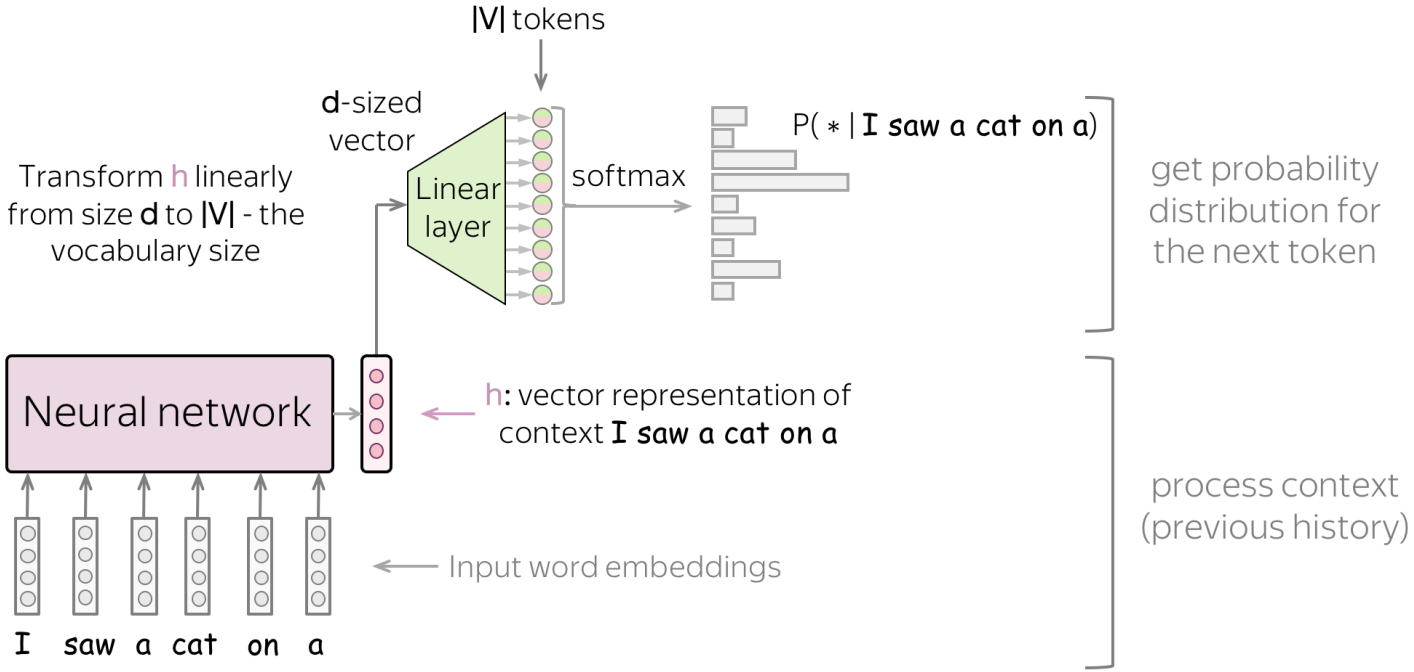




•Efficient Estimation of Word Representations in Vector Space
•Distributed Representations of Words and Phrases and their Compositionality

Transfer Learning with Unlabeled Data

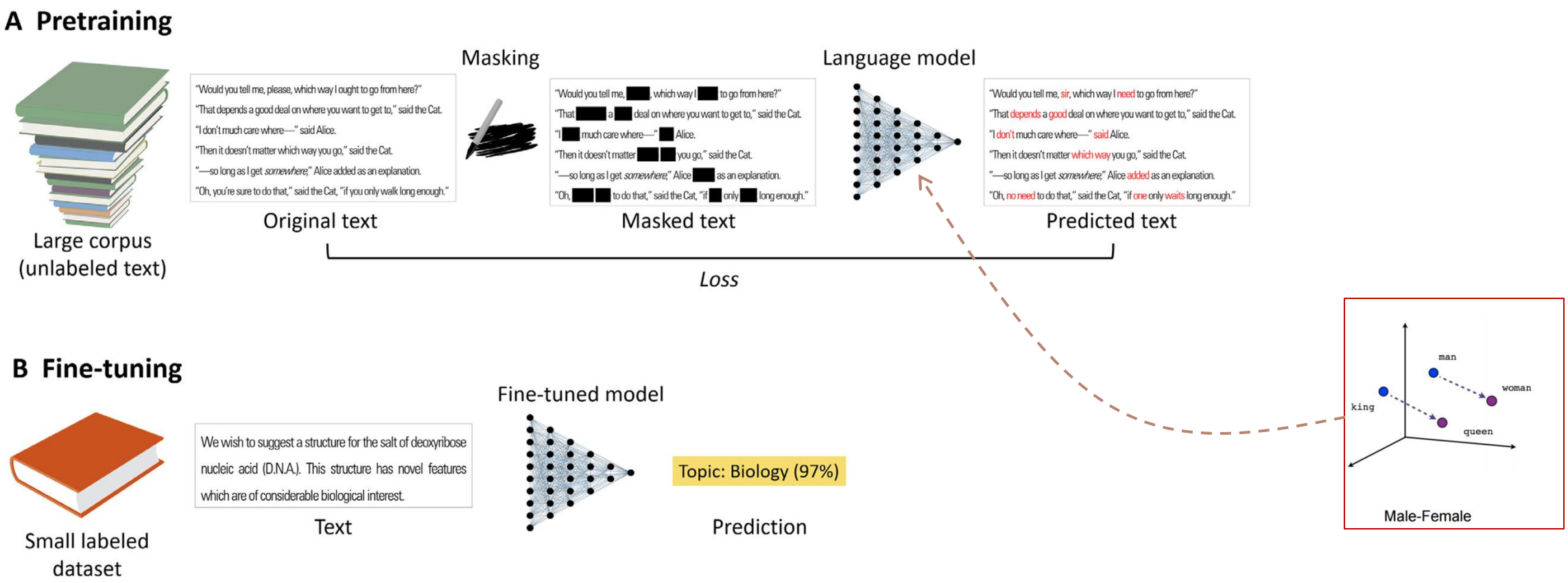
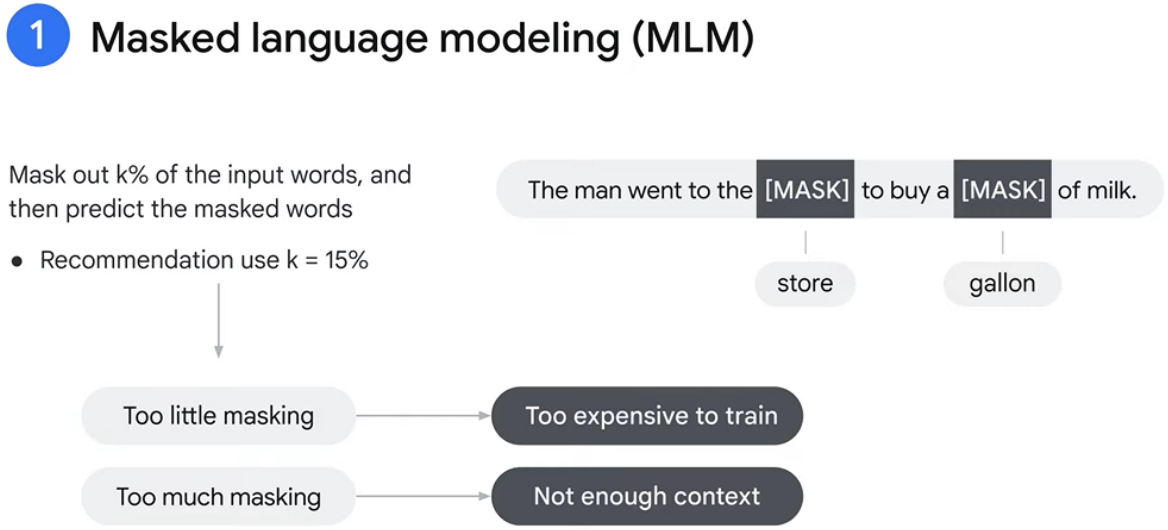


Fig. 2. Language models (A) Language models are trained on self-supervised tasks over huge corpuses of unlabeled text. For example, in the masked language task, some fraction of the tokens in the original text are masked at random, and the language model attempts to predict the original text. (B) (Pre-)trained language models are commonly fine-tuned on downstream tasks over labeled text, through a standard supervised-learning approach. Fine-tuning is typically much faster and provides superior performance than training a model from scratch, especially when labeled data is scarce.

https://pubmed.ncbi.nlm.nih.gov/33897979/

Pretrained Language Model -- BERT

- Source domain data: large-scale unlabeled documents
- Tasks: 1) Masked Language Model and 2) Next Sentence Prediction



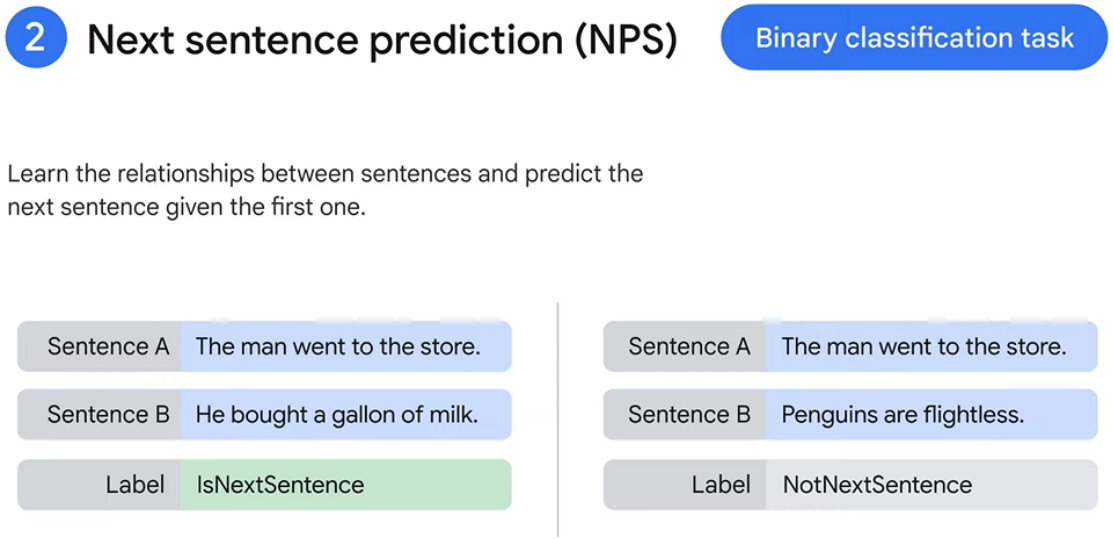


Fig from https://www.archishman.com/post/transformer-models-bert-model-overview

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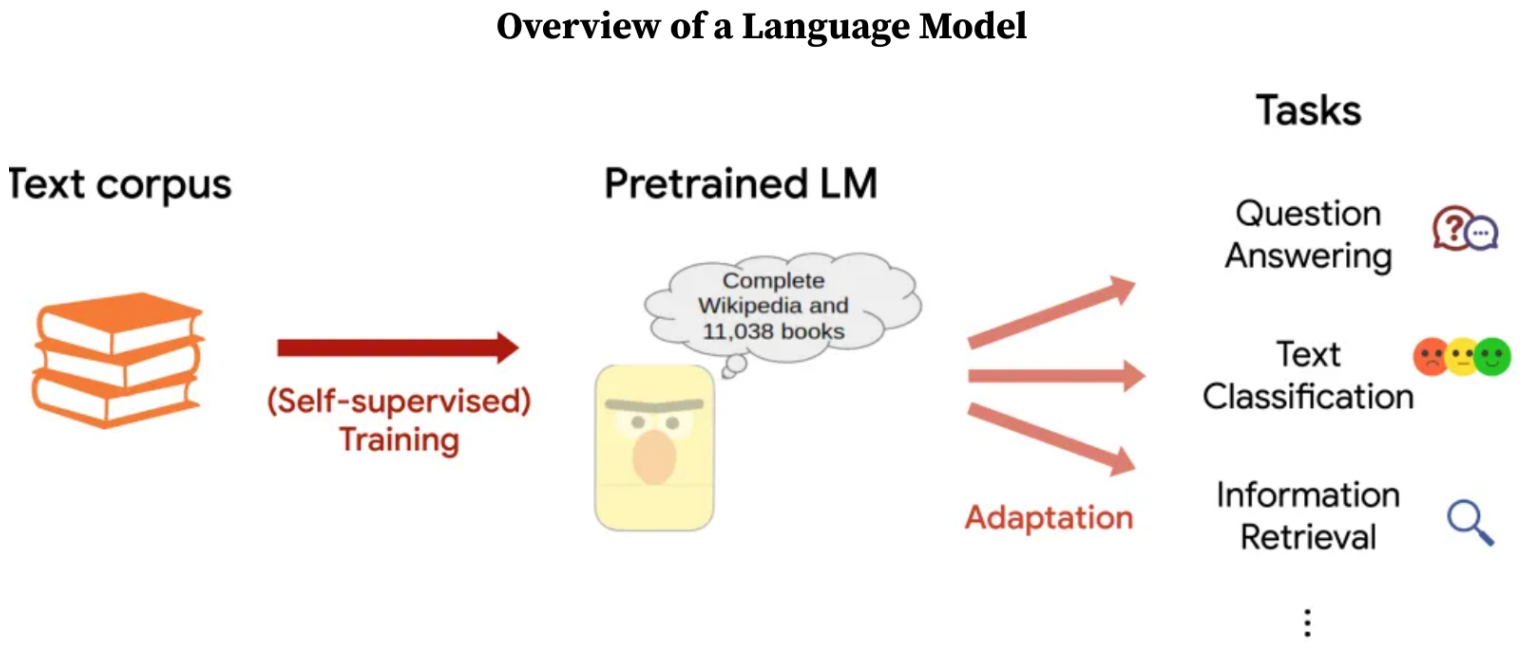
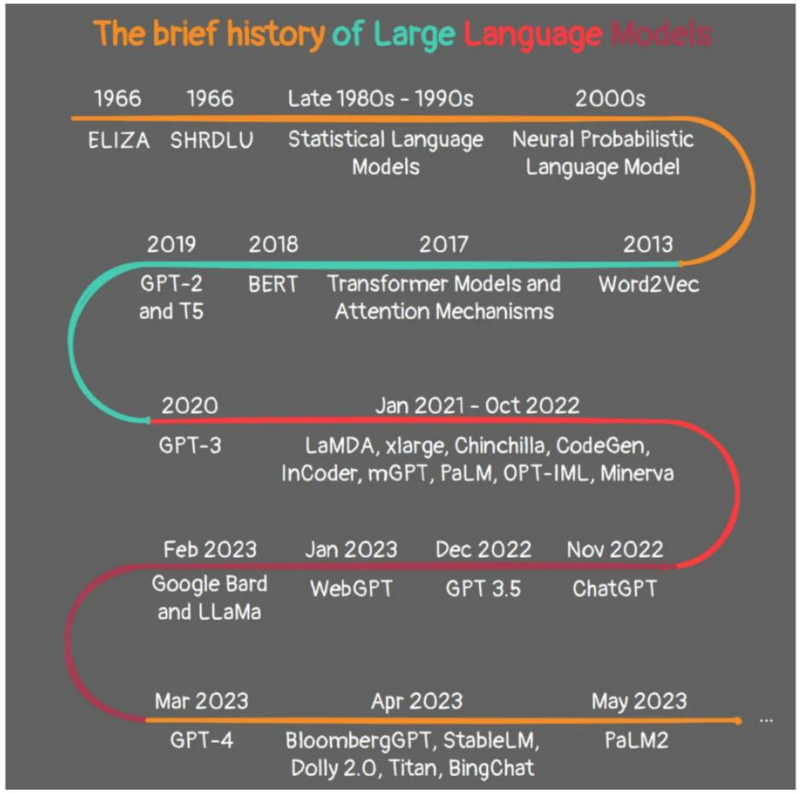


Title: Overview of Language Model
Source: LinkBERT: Improving Language Model Training by Michihiro Yasunaga, Stanford.edu



Title: Timeline of the evolution of different language models

References

- Papers:
  - Jason Yosinski, Jeff Clune, Yoshua Bengio, Hod Lipson, “How transferable are features in deep neural networks?”, NIPS, 2014
  - Yaroslav Ganin, Victor Lempitsky, Unsupervised Domain Adaptation by Backpropagation, ICML, 2015
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- Online:
  - https://www.youtube.com/watch?v=yofjFQddwHE&ab_channel=DeepLearningAI
  - https://www.youtube.com/watch?v=qD6iD4TFsdQ&ab_channel=Hung-yiLee
  - https://www.youtube.com/watch?v=UkQ2FVpDxHg&t=693s&ab_channel=ShusenWang